

OSCAT  
Revision History

| Release | Modul name        | Rev | Date     | History                                                                                                                                                                |
|---------|-------------------|-----|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 274     | F_POLY            | 1.0 | 20.05.08 | New Module                                                                                                                                                             |
| 274     | SEL2_OF_3B        | 1.0 | 19.05.08 | New Module                                                                                                                                                             |
| 274     | ALARM_2           | 1.0 | 19.05.08 | New Module                                                                                                                                                             |
| 274     | RDM               | 1.7 | 18.05.08 | changed constant E to E1                                                                                                                                               |
| 274     | SEL2_OF_3         | 1.0 | 18.05.08 | New Module                                                                                                                                                             |
| 274     | RDMDW             | 1.1 | 18.05.08 | changed constant E to E1                                                                                                                                               |
| 274     | SCALE_R           | 1.0 | 18.05.08 | New Module                                                                                                                                                             |
| 274     | SCALE_D           | 1.0 | 18.05.08 | New Module                                                                                                                                                             |
| 274     | SCALE_B           | 1.0 | 18.05.08 | New Module                                                                                                                                                             |
| 274     | GLOBAL CONSTANTS  |     | 18.05.08 | added charname lists<br>renamed E to E1                                                                                                                                |
| 274     | SCALE             | 1.0 | 18.05.08 | New Module                                                                                                                                                             |
| 274     | REPLACE_CHARS     | 1.0 | 14.05.08 | New Module                                                                                                                                                             |
| 274     | CHARNAME          | 1.0 | 14.05.08 | New Module                                                                                                                                                             |
| 274     | CHARCODE          | 1.0 | 14.05.08 | New Module                                                                                                                                                             |
| 274     | BUFFER_TO_STRING  | 1.2 | 13.05.08 | changed type of pointer to array[0..32767]<br>changed size of string to STRING_LENGTH                                                                                  |
| 274     | BUFFER_SEARCH     | 1.2 | 13.05.08 | changed type of pointer to array[0..32767]<br>changed size of string to STRING_LENGTH                                                                                  |
| 274     | _STRING_TO_BUFFER | 1.2 | 13.05.08 | changed type of pointer to array[0..32767]<br>changed size of string to STRING_LENGTH                                                                                  |
| 274     | _BUFFER_INSERT    | 1.2 | 13.05.08 | changed type of pointer to array[0..32767]<br>changed size of string to STRING_LENGTH                                                                                  |
| 274     | HTML_DECODE       | 1.0 | 03.05.08 | New Module                                                                                                                                                             |
| 274     | HTML_ENCODE       | 1.0 | 03.05.08 | New Module                                                                                                                                                             |
| 274     | GOLD              | 1.0 | 27.04.08 | New Module                                                                                                                                                             |
| 274     | AGDF              | 1.0 | 27.04.08 | New Module                                                                                                                                                             |
| 274     | GDF               | 1.0 | 27.04.08 | New Module                                                                                                                                                             |
| 274     | GLOBAL CONSTANTS  |     | 27.04.08 | added new constant PI05 and PI025                                                                                                                                      |
| 274     | PERIOD2           | 1.0 | 27.04.08 | New Module                                                                                                                                                             |
| 274     | URL_DECODE        | 1.0 | 27.04.08 | New Module                                                                                                                                                             |
| 274     | URL_ENCODE        | 1.0 | 27.04.08 | New Module                                                                                                                                                             |
| 274     | IS_URLCHR         | 1.0 | 27.04.08 | New Module                                                                                                                                                             |
| 274     | WORD_OF_BYTE      | 1.3 | 23.04.08 | reverse order of inputs to be more logical                                                                                                                             |
| 274     | DWORD_OF_BYTE     | 1.2 | 23.04.08 | reverse order of inputs to be more logical                                                                                                                             |
| 274     | IP4_TO_STRING     | 1.0 | 22.04.08 | New Module                                                                                                                                                             |
| 274     | AIN1              | 1.2 | 22.04.08 | corrected error in formula when code_min was set<br>corrected error when sign bit was used<br>optimized code for better performance<br>new functionality overflow_code |
| 274     | CLOG              | 1.1 | 20.04.08 | use ATAN2 instead of ATAN                                                                                                                                              |
| 274     | CARG              | 1.1 | 20.04.08 | use ATAN2 instead of ATAN                                                                                                                                              |
| 274     | ATAN2             | 1.0 | 20.04.08 | New Module                                                                                                                                                             |
| 274     | GLOBAL CONSTANTS  |     | 19.04.08 | added Setup Parameter NETWORK_BUFFER_SIZE                                                                                                                              |
| 274     | DCF77             | 1.9 | 19.04.08 | added input dsi to allow to set daylight savings time when SDT is TRUE<br>added asynchronous SET input                                                                 |
| 274     | URL_TO_STRING     | 1.0 | 18.04.08 | New Module                                                                                                                                                             |
| 274     | STRING_TO_URL     | 1.0 | 18.04.08 | New Module                                                                                                                                                             |
| 274     | REAL_TO_DW        | 1.0 | 18.04.08 | New Module                                                                                                                                                             |
| 274     | DW_TO_REAL        | 1.0 | 18.04.08 | New Module                                                                                                                                                             |
| 274     | GAUSSCD           | 1.0 | 07.04.08 | New Module                                                                                                                                                             |
| 274     | ERFC              | 1.0 | 07.04.08 | New Module                                                                                                                                                             |
| 274     | ERF               | 1.0 | 07.04.08 | New Module                                                                                                                                                             |
| 274     | GLOBAL CONSTANTS  |     | 07.04.08 | deleted unused setup parameter STEP7                                                                                                                                   |
| 274     | YEAR_BEGIN        | 1.2 | 07.04.08 | deleted unused step7 code                                                                                                                                              |
| 274     | YEAR              | 1.3 | 07.04.08 | deleted unused step7 code                                                                                                                                              |
| 274     | WEEKDAY           | 1.3 | 07.04.08 | deleted unused step7 code                                                                                                                                              |
| 274     | SET_DATE          | 1.6 | 07.04.08 | deleted unused step7 code                                                                                                                                              |
| 274     | HOLIDAY           | 1.4 | 07.04.08 | improved algorithm for higher speed                                                                                                                                    |
| 274     | EASTER            | 1.3 | 07.04.08 | improved algorithm for higher speed                                                                                                                                    |
| 274     | DAY_OF_DATE       | 1.3 | 07.04.08 | deleted unused step7 code                                                                                                                                              |
| 274     | GAUSS             | 1.0 | 06.04.08 | New Module                                                                                                                                                             |
| 274     | MODR              | 1.3 | 04.04.08 | added type conversion to avoid warnings under codesys 3.0                                                                                                              |
| 274     | IS_SORTED         | 1.1 | 04.04.08 | added type conversion to avoid warnings under codesys 3.0                                                                                                              |

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| 274 | FLOOR2           | 1.1 | 04.04.08 | added type conversion to avoid warnings under codesys 3.0                                                      |
| 274 | CEIL2            | 1.1 | 04.04.08 | added type conversion to avoid warnings under codesys 3.0                                                      |
| 274 | ARRAY_HAV        | 1.0 | 02.04.08 | New Module                                                                                                     |
| 274 | ARRAY_GAV        | 1.0 | 02.04.08 | New Module                                                                                                     |
| 274 | _ARRAY_MUL       | 1.0 | 02.04.08 | New Module                                                                                                     |
| 274 | _ARRAY_ADD       | 1.0 | 02.04.08 | New Module                                                                                                     |
| 274 | _ARRAY_ABS       | 1.0 | 02.04.08 | New Module                                                                                                     |
| 273 | _ARRAY_SHUFFLE   | 1.2 | 30.03.08 | changed last in rdm2 from pos to i+pos for more randomness                                                     |
| 273 | TO_UML           | 1.0 | 29.03.08 | New Module                                                                                                     |
| 273 | UPPERCASE        | 1.3 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | TRIM1            | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | TRIM             | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | TICKER           | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | REPLACE_UML      | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)<br>new code to avoid pointer out of range<br>use new function to_uml   |
| 273 | REPLACE_ALL      | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | REAL_TO_STRF     | 1.5 | 29.03.08 | changed STRING to STRING(20)                                                                                   |
| 273 | MIRROR           | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | MESSAGE          | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | LOWERCASE        | 1.3 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | IS_UPPER         | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | IS_NUM           | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | IS_NCC           | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | IS_LOWER         | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | IS_HEX           | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | IS_CTRL          | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | IS_CC            | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | IS_ALPHA         | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | IS_ALNUM         | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | HEX_TO_DWORD     | 1.3 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | FINDP            | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | FINDB_NUM        | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | FINDB_NONUM      | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | FINDB            | 1.3 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | FIND_NUM         | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | FIND_NONUM       | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | FIND_CTRL        | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | FIND_CHAR        | 1.2 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | EXEC             | 1.4 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | DWORD_TO_STRH    | 1.3 | 29.03.08 | changed STRING to STRING(8)                                                                                    |
| 273 | DWORD_TO_STRF    | 1.1 | 29.03.08 | changed STRING to STRING(20)<br>limit the output string to max 20 digits                                       |
| 273 | DWORD_TO_STRB    | 1.2 | 29.03.08 | changed STRING to STRING(32)                                                                                   |
| 273 | COUNT_CHAR       | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | CODE             | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | CHR              | 1.2 | 29.03.08 | changed STRING to STRING(1)                                                                                    |
| 273 | CAPITALIZE       | 1.1 | 29.03.08 | changed STRING to STRING(STRING_LENGTH)                                                                        |
| 273 | BYTE_TO_STRH     | 1.3 | 29.03.08 | changed STRING to STRING(2)                                                                                    |
| 273 | BYTE_TO_STRB     | 1.2 | 29.03.08 | changed STRING to STRING(8)                                                                                    |
| 273 | GLOBAL CONSTANTS |     | 28.03.08 | Added setup parameter STRING_LENGTH                                                                            |
| 273 | IS_SORTED        | 1.0 | 28.03.08 | New Module                                                                                                     |
| 273 | TRIME            | 1.0 | 28.03.08 | New Module                                                                                                     |
| 273 | FIX              | 1.0 | 28.03.08 | New Module                                                                                                     |
| 273 | FILL             | 1.0 | 28.03.08 | New Module                                                                                                     |
| 273 | FIND_CTRL        | 1.1 | 26.03.08 | character 127 is now recognized as control                                                                     |
| 273 | FIND_CHAR        | 1.1 | 26.03.08 | char will now accept extended ascii                                                                            |
| 273 | FIB              | 1.3 | 26.03.08 | function now returns -1 for input < 0 or > 46                                                                  |
| 273 | METER            | 1.2 | 24.03.08 | use data_type real2 to extend accuracy to 15 digits total<br>do not use ft_int which adds unnecessary overhead |
| 273 | PERIOD           | 1.3 | 22.03.08 | function would deliver wrong results when d1 and dx are a leap_year                                            |
| 273 | DAYS_DELTA       | 1.2 | 22.03.08 | changed output from int to dint because the total date range is 49710 days                                     |
| 273 | DAY_OF_DATE      | 1.2 | 22.03.08 | changed output from int to dint because the total date range is 49710 days                                     |
| 273 | DATE_ADD         | 1.3 | 22.03.08 | fixed some bugs when month was negative                                                                        |
| 273 | R2_ABS           | 1.0 | 21.03.08 | New Module                                                                                                     |

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| 273 | FRACT             | 1.2 | 21.03.08 | Replaced TRUNC with D_TRUNC for compatibility reasons                                                                        |
| 273 | MODR              | 1.2 | 21.03.08 | Replaced TRUNC with D_TRUNC for compatibility reasons                                                                        |
| 273 | FLOOR2            | 1.0 | 21.03.08 | New Module                                                                                                                   |
| 273 | CEIL              | 1.1 | 21.03.08 | Replaced TRUNC with real_to_int for compatibility reasons                                                                    |
| 273 | EVEN              | 1.1 | 21.03.08 | Changed type of input IN from int to dint                                                                                    |
| 273 | FLOOR             | 1.1 | 21.03.08 | Replaced TRUNC with real_to_int for compatibility reasons                                                                    |
| 273 | CEIL2             | 1.0 | 21.03.08 | New Module                                                                                                                   |
| 273 | D_TRUNC           | 1.0 | 21.03.08 | New Module                                                                                                                   |
| 273 | R2_ADD2           | 1.0 | 20.03.08 | New Module                                                                                                                   |
| 273 | R2_MUL            | 1.0 | 20.03.08 | New Module                                                                                                                   |
| 273 | R2_ADD            | 1.0 | 20.03.08 | New Module                                                                                                                   |
| 273 | TRIM1             | 1.1 | 20.03.08 | avoid to call replace with pos = 0 because some systems will produce an error                                                |
| 273 | TRIM              | 1.1 | 20.03.08 | avoid to call replace with pos = 0 because some systems will produce an error                                                |
| 273 | REAL_TO_STRF      | 1.4 | 20.03.08 | changed trunc to dint because trunc was generating wrong values on wago 842<br>corrected an error with negative numbers      |
| 273 | EXEC              | 1.3 | 20.03.08 | Make sure output of exec will always be in the format x.y                                                                    |
| 273 | IS_NCC            | 1.0 | 19.03.08 | New Module                                                                                                                   |
| 273 | IS_CC             | 1.0 | 19.03.08 | New Module                                                                                                                   |
| 273 | CRC_CHECK         | 1.1 | 16.03.08 | changed type of input size to uint<br>improved code for better performance                                                   |
| 273 | CRC_GEN           | 1.4 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | PARSET            | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | HEAT_METER        | 1.3 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | _STRING_TO_BUFFER | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | BUFFER_SEARCH     | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | BUFFER_TO_STRING  | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | _BUFFER_INSERT    | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | GEN_SIN           | 1.4 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | GEN_RDT           | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | FT_PT1            | 1.8 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | FT_INT            | 1.9 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | DST               | 1.3 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0<br>code improvement for better performance                         |
| 273 | ESR_MON_R4        | 1.3 | 16.03.08 | deleted wrong conversion real_to_dword                                                                                       |
| 273 | ARRAY_VAR         | 1.2 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | ARRAY_TREND       | 1.2 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | ARRAY_SUM         | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | ARRAY_SPR         | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | ARRAY_SDV         | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | ARRAY_MIN         | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | ONTIME            | 1.5 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | ARRAY_MAX         | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | ARRAY_AVG         | 1.2 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | _ARRAY_SORT       | 1.6 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | _ARRAY_SHUFFLE    | 1.1 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | _ARRAY_MEDIAN     | 1.5 | 16.03.08 | changed type of input size to uint<br>improved performance                                                                   |
| 273 | _ARRAY_INIT       | 1.3 | 16.03.08 | changed type of input size to uint                                                                                           |
| 273 | SET_DATE          | 1.5 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | SET_DT            | 1.5 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | SUN_POS           | 1.3 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | TC_S              | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | TUNE              | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | TUNE2             | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0<br>improved performance                                            |
| 273 | WORK_WEEK         | 1.3 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | DCF77             | 1.8 | 16.03.08 | Changed output weekday to wday and dst to ds for compatibility reasons                                                       |
| 273 | BIT_LOAD_DW       | 1.1 | 16.03.08 | Changed input bit to val for compatibility reasons                                                                           |
| 273 | BIT_LOAD_W        | 1.1 | 16.03.08 | Changed input bit to val for compatibility reasons                                                                           |
| 273 | BIT_LOAD_B        | 1.1 | 16.03.08 | Changed input bit to val for compatibility reasons                                                                           |
| 273 | AIN1              | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | AIN               | 1.5 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0                                                                    |
| 273 | _BUFFER_CLEAR     | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0<br>changed type of input size to uint<br>deleted unused variable i |

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| 273 | _BUFFER_INIT      | 1.1 | 16.03.08 | added type conversion to avoid warnings under codesys 3.0<br>changed type of input size to uint      |
| 273 | BYTE_TO_BITS      | 1.1 | 16.03.08 | renamed from byte_to_bit to byte_to_bits for compatibility reasons                                   |
| 273 | BYTE_TO_BIT       | 1.1 | 16.03.08 | renamed from byte_to_bit to byte_to_bits for compatibility reasons                                   |
| 273 | SET_TOD           | 1.5 | 16.03.08 | added type conversion to avoid warning under codesys 3.0                                             |
| 273 | RDM               | 1.6 | 16.03.08 | added type conversion to avoid warning under codesys 3.0                                             |
| 273 | ROUND             | 1.4 | 16.03.08 | added type conversion to avoid warning under codesys 3.0                                             |
| 273 | ASINH             | 1.2 | 16.03.08 | changed sequence of calculations to improve performance                                              |
| 273 | ACOSH             | 1.2 | 16.03.08 | changed sequence of calculations to improve performance                                              |
| 273 | DIFFER            | 1.1 | 16.03.08 | Improved code for higher speed                                                                       |
| 272 | RDMDW             | 1.0 | 14.03.08 | New Module                                                                                           |
| 272 | MULTIME           | 1.1 | 14.03.08 | rounded the result after the last digit                                                              |
| 272 | HOUR_TO_TOD       | 1.1 | 14.03.08 | rounded the input after the last digit                                                               |
| 272 | HOUR_TO_TIME      | 1.1 | 14.03.08 | rounded the input after the last digit                                                               |
| 272 | MINUTE_TO_TIME    | 1.1 | 14.03.08 | rounded the input after the last digit                                                               |
| 272 | SECOND_TO_TIME    | 1.1 | 14.03.08 | rounded the input after the last digit                                                               |
| 272 | SET_TOD           | 1.4 | 14.03.08 | changed code to avoid rounding problem at last digit of millisecond                                  |
| 272 | TC_S              | 1.0 | 13.03.08 | New Module                                                                                           |
| 272 | TC_MS             | 1.0 | 13.03.08 | New Module                                                                                           |
| 272 | TC_US             | 1.0 | 13.03.08 | New Module                                                                                           |
| 272 | SRAMP             | 1.0 | 13.03.08 | New Module                                                                                           |
| 272 | FT_INT            | 1.8 | 13.03.08 | changed preset value out_min to -1e37<br>stop integrator at the limits<br>optimized code             |
| 272 | CMP               | 1.0 | 12.03.08 | New Module                                                                                           |
| 272 | SINH              | 1.3 | 11.03.08 | improved accuracy around 0                                                                           |
| 272 | SINC              | 1.0 | 11.03.08 | New Module                                                                                           |
| 272 | LANGEVIN          | 1.1 | 11.03.08 | changed formula to avoid problems when x = 0                                                         |
| 272 | SQRTN             | 1.2 | 11.03.08 | added result 0 for compatability reasons                                                             |
| 272 | FRACT             | 1.1 | 11.03.08 | added dint_to_real for compatability reasons<br>now returns 0 for number > 2e9<br>changed input to x |
| 272 | RND               | 1.0 | 11.03.08 | New Module                                                                                           |
| 272 | ROUND             | 1.3 | 11.03.08 | corrected an error with negative numbers                                                             |
| 272 | SIGMOID           | 1.1 | 11.03.08 | extended range of valid inputs to +inv / -inv                                                        |
| 272 | FIB               | 1.2 | 10.03.08 | changed output to dint instead of real                                                               |
| 272 | FACT              | 1.3 | 10.03.08 | changed output of fact to dint to allow bigger values                                                |
| 272 | COTH              | 1.3 | 10.03.08 | extended range of valid inputs to +/- INV<br>changed formula for better accuracy                     |
| 272 | TANH              | 1.2 | 10.03.08 | corrected an error in formula<br>extended range of valid inputs to +/- INV                           |
| 272 | RDM               | 1.5 | 10.03.08 | make sure last will be between 0 and 1 to avoid invalid results                                      |
| 272 | _BUFFER_INSERT    | 1.0 | 09.03.08 | New Module                                                                                           |
| 272 | CODE              | 1.0 | 09.03.08 | New Module                                                                                           |
| 272 | HEAT_METER        | 1.2 | 06.03.08 | corrected an error in pulse mode                                                                     |
| 272 | ISC_UPPER         | 1.0 | 06.03.08 | New Module                                                                                           |
| 272 | ISC_LOWER         | 1.0 | 06.03.08 | New Module                                                                                           |
| 272 | ISC_HEX           | 1.0 | 06.03.08 | New Module                                                                                           |
| 272 | ISC_CTRL          | 1.0 | 06.03.08 | New Module                                                                                           |
| 272 | ISC_NUM           | 1.0 | 06.03.08 | New Module                                                                                           |
| 272 | ISC_ALPHA         | 1.0 | 06.03.08 | New Module                                                                                           |
| 272 | LOWERCASE         | 1.2 | 06.03.08 | added support for exteded Ascii                                                                      |
| 272 | UPPERCASE         | 1.2 | 06.03.08 | added support for exteded Ascii                                                                      |
| 272 | TO_LOWER          | 1.0 | 06.03.08 | New Module                                                                                           |
| 272 | TO_UPPER          | 1.0 | 06.03.08 | New Module                                                                                           |
| 272 | GLOBAL CONSTANTS  |     | 06.03.08 | Added EXTENDED_ASCII                                                                                 |
| 272 | BUFFER_TO_STRING  | 1.0 | 05.03.08 | New Module                                                                                           |
| 272 | BUFFER_SEARCH     | 1.0 | 05.03.08 | New Module                                                                                           |
| 272 | _STRING_TO_BUFFER | 1.0 | 05.03.08 | New Module                                                                                           |
| 272 | _BUFFER_INIT      | 1.0 | 05.03.08 | New Module                                                                                           |
| 272 | _BUFFER_CLEAR     | 1.0 | 05.03.08 | New Module                                                                                           |
| 272 | CAPITALIZE        | 1.0 | 04.03.08 | New Module                                                                                           |
| 272 | IP4_CHECK         | 1.0 | 04.03.08 | New Module                                                                                           |
| 272 | REAL_TO_STRF      | 1.3 | 04.03.08 | result is now rounded instead of trunc<br>corrected an error with reals < 1                          |

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|-----|----------------|-----|----------|------------------------------------------------------------|
| 272 | _ARRAY_SHUFFLE | 1.0 | 04.03.08 | New Module                                                 |
| 272 | HEX_TO_DWORD   | 1.2 | 04.03.08 | added support for a..f and return 0 for invalid string     |
| 272 | EXEC           | 1.2 | 04.03.08 | add a 0 to the string if a '.' is at the end of the string |
| 272 | REPLACE_UML    | 1.0 | 29.02.08 | New Module                                                 |
| 272 | IS_UPPER       | 1.0 | 29.02.08 | New Module                                                 |
| 272 | IS_NUM         | 1.0 | 29.02.08 | New Module                                                 |
| 272 | IS_LOWER       | 1.0 | 29.02.08 | New Module                                                 |
| 272 | IS_IP4         | 1.0 | 29.02.08 | New Module                                                 |
| 272 | IS_HEX         | 1.0 | 29.02.08 | New Module                                                 |
| 272 | IS_CTRL        | 1.0 | 29.02.08 | New Module                                                 |
| 272 | IS_ALPHA       | 1.0 | 29.02.08 | New Module                                                 |
| 272 | IS_ALNUM       | 1.0 | 29.02.08 | New Module                                                 |
| 272 | IP4_DECODE     | 1.0 | 29.02.08 | New Module                                                 |
| 272 | COUNT_CHAR     | 1.0 | 29.02.08 | New Module                                                 |
| 272 | FINDB_NUM      | 1.1 | 29.02.08 | Improved code for higher speed                             |
| 272 | FINDB_NONUM    | 1.1 | 29.02.08 | Improved code for higher speed                             |
| 272 | FIND_CHAR      | 1.0 | 29.02.08 | New Module                                                 |
| 272 | FIND_CTRL      | 1.0 | 29.02.08 | New Module                                                 |
| 272 | FINDP          | 1.1 | 29.02.08 | ADDED MAX(pos,1) in loop initialization                    |
| 272 | BIT_LOAD_B     | 1.0 | 29.02.08 | New Module                                                 |
| 272 | BIT_LOAD_DW    | 1.0 | 29.02.08 | New Module                                                 |
| 272 | BIT_LOAD_W     | 1.0 | 29.02.08 | New Module                                                 |
| 272 | RDM2           | 1.0 | 29.02.08 | New Module                                                 |
| 272 | FINDB          | 1.2 | 29.02.08 | added findb := 0 for compatibility reasons                 |
| 272 | FIND_NUM       | 1.1 | 29.02.08 | added input pos to start search at position                |
| 272 | FIND_NONUM     | 1.1 | 29.02.08 | added input pos to start search at position                |
| 271 | AOUT           | 1.4 | 23.02.08 | Improved code for higher speed                             |
| 271 | AOUT1          | 1.0 | 23.02.08 | New Module                                                 |
| 271 | AIN1           | 1.0 | 23.02.08 | New Module                                                 |
| 271 | POLYNOM_INT    | 1.2 | 22.02.08 | Improved code for higher speed                             |
| 271 | TRIANGLE_A     | 1.1 | 22.02.08 | Improved code for higher speed                             |
| 271 | SPHERE_V       | 1.2 | 22.02.08 | Improved code for higher speed                             |
| 271 | CONE_V         | 1.2 | 22.02.08 | Improved code for higher speed                             |
| 271 | CIRCLE_C       | 1.1 | 22.02.08 | Improved code for higher speed                             |
| 271 | CIRCLE_A       | 1.2 | 22.02.08 | Improved code for higher speed                             |
| 271 | CABS           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CACOS          | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CACOSH         | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CADD           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CARG           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CASIN          | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CASINH         | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CATAN          | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CATANH         | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CCON           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CCOS           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CCOSH          | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CDIV           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CEXP           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CINV           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CLOG           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CMUL           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CPOL           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CPOW           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CSET           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CSIN           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CSINH          | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CSQRT          | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CSUB           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CTAN           | 1.0 | 21.02.08 | New Module                                                 |
| 271 | CTANH          | 1.0 | 21.02.08 | New Module                                                 |
| 271 | HYPOT          | 1.0 | 21.02.08 | New Module                                                 |
| 271 | TEMPERATURE    | 1.0 | 21.02.08 | New Module                                                 |
| 271 | RTC_2          | 1.1 | 20.02.08 | Added millisecond set input                                |
| 271 | RTC_MS         | 1.1 | 20.02.08 | Added millisecond set input                                |

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|     |                |     |          |                                                                                                                                             |
|-----|----------------|-----|----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 271 | WORD_OF_BYTE   | 1.2 | 19.02.08 | replaced + with or for better compatibility                                                                                                 |
| 271 | D_TRIG         | 1.1 | 19.02.08 | Improved code for higher speed                                                                                                              |
| 271 | STACK_32       | 1.1 | 19.02.08 | Improved code for higher speed                                                                                                              |
| 271 | STACK_16       | 1.1 | 19.02.08 | Improved code for higher speed                                                                                                              |
| 271 | FIFO_32        | 1.1 | 19.02.08 | Improved code for higher speed                                                                                                              |
| 271 | FIFO_16        | 1.1 | 19.02.08 | Improved code for higher speed                                                                                                              |
| 271 | PARITY         | 1.3 | 18.02.08 | Improved code for higher speed                                                                                                              |
| 271 | CHECK_PARITY   | 1.3 | 18.02.08 | Improved code for higher speed                                                                                                              |
| 271 | BYTE_OF_BIT    | 1.1 | 18.02.08 | Improved code for higher speed                                                                                                              |
| 271 | REVERSE        | 1.1 | 18.02.08 | Improved code for higher speed                                                                                                              |
| 271 | BLIND_SECURITY | 1.2 | 12.02.08 | added config variables rain_up and wind_up<br>position for rain and wind is now configurable                                                |
| 27  | FT_INT         | 1.7 | 08.02.08 | Added interim variable to allow for changes smaller then 8 digits of output                                                                 |
| 27  | METER          | 1.1 | 08.02.08 | Use new version of ft_int and deleted unnecessary limits for integrators                                                                    |
| 27  | SWAP_BYTE2     | 1.0 | 04.02.08 | New Module                                                                                                                                  |
| 27  | SWAP_BYTE      | 1.0 | 04.02.08 | New Module                                                                                                                                  |
| 27  | REPLACE_ALL    | 1.0 | 04.02.08 | New Module                                                                                                                                  |
| 27  | CHR            | 1.1 | 04.02.08 | return string would not be terminated properly                                                                                              |
| 27  | FINDP          | 1.0 | 04.02.08 | New Module                                                                                                                                  |
| 27  | MIRROR         | 1.0 | 04.02.08 | New Module                                                                                                                                  |
| 27  | TRIM1          | 1.0 | 04.02.08 | New Module                                                                                                                                  |
| 27  | LOWERCASE      | 1.1 | 04.02.08 | improved performance<br>added support for Ä,Ö,Ü                                                                                             |
| 27  | UPPERCASE      | 1.1 | 04.02.08 | improved performance<br>added support for Ä,Ö,Ü                                                                                             |
| 27  | EASTER         | 1.2 | 03.02.08 | Modified code for better performance                                                                                                        |
| 27  | RDM            | 1.4 | 02.02.08 | changed algorithm to avoid non iec functions and guarantee more randomness                                                                  |
| 27  | BLIND_SCENE    | 1.1 | 29.01.08 | Replaced shr/shl with and for better performance                                                                                            |
| 27  | FT_RMP         | 1.4 | 25.01.08 | Improved algorithm for higher speed<br>allow KR and KF to be 0                                                                              |
| 27  | DCF77          | 1.7 | 23.01.08 | added sdt input which is used to initialize rtc and rtc1 during first cycle.                                                                |
| 27  | RTC_2          | 1.0 | 21.01.08 | New module                                                                                                                                  |
| 27  | RTC_MS         | 1.0 | 20.01.08 | New Module                                                                                                                                  |
| 27  | SUN_POS        | 1.2 | 18.01.08 | further performance improvements<br>only calculate once every 10 seconds                                                                    |
| 27  | SUN_TIME       |     | 17.01.08 | calculation is now only performed once a day                                                                                                |
| 27  | TIMER_1        | 1.6 | 17.01.08 | New code with better performance<br>Output is now only active during the time of day it is supposed to<br>no pulse is generated at power up |
| 27  | COUNT_BR       | 1.0 | 16.01.08 | New Module                                                                                                                                  |
| 27  | SELECT_8       | 1.0 | 16.01.08 | New Module                                                                                                                                  |
| 27  | INC            | 1.1 | 15.01.08 | Allow for negative increment                                                                                                                |
| 27  | RANGE_TO_WORD  | 1.0 | 09.01.08 | New Module                                                                                                                                  |
| 27  | RANGE_TO_BYTE  | 1.0 | 09.01.08 | New Module                                                                                                                                  |
| 27  | WORD_TO_RANGE  | 1.0 | 09.01.08 | New Module                                                                                                                                  |
| 27  | BYTE_TO_RANGE  | 1.0 | 09.01.08 | New Module                                                                                                                                  |
| 27  | DEW_TEMP       | 1.4 | 08.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | LEAP_OF_DATE   | 1.2 | 08.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | DAY_OF_MONTH   | 1.4 | 08.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | COTH           | 1.2 | 08.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | ROUND          | 1.2 | 08.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | SH_2           | 1.4 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | STAIR          | 1.2 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | SCALE_B8       | 1.2 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | SCALE_B4       | 1.2 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | SCALE_B2       | 1.2 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | GEN_SQR        | 1.3 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | GEN_SIN        | 1.3 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | TEMP_SI        | 1.4 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | TEMP_NI        | 1.4 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | BAR_GRAPH      | 1.2 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | KMH_TO_MS      | 1.1 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | F_TO_C         | 1.1 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | SUN_POS        | 1.1 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |
| 27  | SUN_TIME       | 1.3 | 06.01.08 | Improved algorithm for higher speed                                                                                                         |

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|    |                |     |          |                                                                              |
|----|----------------|-----|----------|------------------------------------------------------------------------------|
| 27 | _ARRAY_SORT    | 1.5 | 05.01.08 | Improved algorithm for higher speed                                          |
| 27 | SINH           | 1.2 | 05.01.08 | Improved algorithm for higher speed                                          |
| 27 | COSH           | 1.2 | 05.01.08 | Improved algorithm for higher speed                                          |
| 27 | ATANH          | 1.1 | 05.01.08 | Improved algorithm for higher speed                                          |
| 27 | ACOTH          | 1.1 | 05.01.08 | Improved algorithm for higher speed                                          |
| 27 | HYST_3         | 1.2 | 05.01.08 | Improved code for higher performance                                         |
| 27 | HYST_2         | 1.1 | 05.01.08 | Improved code for higher performance                                         |
| 27 | FT_PT1         | 1.7 | 05.01.08 | Improved code for higher performance                                         |
| 27 | FT_PID         | 1.6 | 05.01.08 | Improved code for higher performance                                         |
| 27 | FT_PI          | 1.2 | 05.01.08 | Improved performance                                                         |
| 27 | FT_INT         | 1.6 | 05.01.08 | Improved performance                                                         |
| 27 | TEMP_PT        | 1.5 | 05.01.08 | replaced /2 with * 0.5 for better performance                                |
| 27 | RDM            | 1.3 | 05.01.08 | Corrected an error with high values of sps timer                             |
| 27 | PERIOD         | 1.2 | 04.01.08 | Improved performance                                                         |
| 27 | DAY_OF_YEAR    | 1.2 | 04.01.08 | Improved performance                                                         |
| 27 | YEAR_BEGIN     | 1.1 | 04.01.08 | formula for step7 was incorrect during leap years                            |
| 27 | SET_DATE       | 1.4 | 03.01.08 | Improved performance                                                         |
| 27 | IRTRANS_DECODE | 1.0 | 02.01.08 | New Module                                                                   |
| 27 | IRTRANS_8      | 1.0 | 02.01.08 | New Module                                                                   |
| 27 | IRTRANS_4      | 1.0 | 02.01.08 | New Module                                                                   |
| 27 | IRTRANS_1      | 1.0 | 02.01.08 | New Module                                                                   |
| 27 | DRIVER_4       | 1.0 | 02.01.08 | New Module                                                                   |
| 27 | DRIVER_1       | 1.0 | 02.01.08 | New Module                                                                   |
| 27 | CLK_DIV        | 1.1 | 02.01.08 | Improved performance                                                         |
| 27 | BYTE_OF_DWORD  | 1.1 | 02.01.08 | Improved performance                                                         |
| 27 | WORD_OF_DWORD  | 1.1 | 02.01.08 | Improved performance                                                         |
| 27 | FIB            | 1.1 | 02.01.08 | deleted unused variable pt                                                   |
| 27 | WORD_OF_BYTE   | 1.1 | 02.01.08 | Improved performance                                                         |
| 27 | CRC_GEN        | 1.3 | 02.01.08 | Improved performance                                                         |
| 27 | DWORD_OF_BYTE  | 1.1 | 02.01.08 | Improved performance                                                         |
| 26 | BLIND_INPUT    | 1.2 | 30.12.07 | added click_timeout variable and function                                    |
| 26 | FIB            | 1.0 | 27.12.07 | New Module                                                                   |
| 26 | WATER_CP       | 1.1 | 27.12.07 | Increased accuracy                                                           |
| 26 | SHR1           | 1.1 | 27.12.07 | changed code for better performance                                          |
| 26 | SHL1           | 1.1 | 27.12.07 | changed code for better performance                                          |
| 26 | LTCH_4         | 1.1 | 27.12.07 | changed code for better performance                                          |
| 26 | LTCH           | 1.1 | 27.12.07 | changed code for better performance                                          |
| 26 | FF_JKE         | 1.1 | 27.12.07 | changed code for better performance                                          |
| 26 | FF_DRE         | 1.1 | 27.12.07 | changed code for better performance                                          |
| 26 | FF_D4E         | 1.1 | 27.12.07 | changed code for better performance                                          |
| 26 | FF_D2E         | 1.1 | 27.12.07 | changed code for better performance                                          |
| 26 | STAIR          | 1.1 | 27.12.07 | changed code for better performance                                          |
| 26 | STAIR2         | 1.1 | 27.12.07 | changed code for better performance                                          |
| 26 | HYST_3         | 1.1 | 27.12.07 | changed code for better performance                                          |
| 26 | ACTUATOR_2P    | 1.5 | 27.12.07 | changed code for better performance                                          |
| 26 | LINEAR_INT     | 1.1 | 27.12.07 | changed code for better performance                                          |
| 26 | STORE_8        | 1.0 | 26.12.07 | New Module                                                                   |
| 26 | MESSAGE_8      | 1.0 | 26.12.07 | New Module                                                                   |
| 26 | FADE           | 1.0 | 26.12.07 | New Module                                                                   |
| 26 | RMP_B          | 1.6 | 25.12.07 | changed code for better performance                                          |
| 26 | RMP_W          | 1.7 | 25.12.07 | corrected an error while step response was too slow for fast rist times      |
| 26 | HEAT_METER     | 1.1 | 25.12.07 | added config variable return_meter<br>added output C for current consumption |
| 26 | MATRIX         | 1.2 | 23.12.07 | added exit statement in for loop instead of i:=5                             |
| 26 | WEEKDAY        | 1.2 | 23.12.07 | correction for step7                                                         |
| 26 | TEMP_PT        | 1.4 | 23.12.07 | avoid a negative square root if input values are wrong                       |
| 26 | YEAR           | 1.2 | 23.12.07 | changed code for better performance                                          |
| 26 | WORK_WEEK      | 1.2 | 19.12.07 | changed code for better performance<br>changed code to comply with ISO8601   |
| 26 | YEAR_BEGIN     | 1.0 | 19.12.07 | New Module                                                                   |
| 26 | TEMP_NI        | 1.3 | 17.12.07 | changed code for better performance                                          |
| 26 | POLYNOM_INT    | 1.1 | 17.12.07 | init makes no sense for a function                                           |
| 26 | TEMP_SI        | 1.3 | 17.12.07 | changed code for better performance                                          |
| 26 | YEAR           | 1.2 | 16.12.07 | changed code for better performance                                          |
| 26 | SET_DATE       | 1.3 | 16.12.07 | changed code for better performance                                          |

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|     |               |     |          |                                                              |
|-----|---------------|-----|----------|--------------------------------------------------------------|
| 26  | OSCAT_VERSION | 1.1 | 16.12.07 | added possibility to return date and version depending on in |
| 26  | CLK_8MS       |     | 16.12.07 | No longer supported is replaced by CLK_N                     |
| 26  | CLK_4MS       |     | 16.12.07 | No longer supported is replaced by CLK_N                     |
| 26  | CLK_2MS       |     | 16.12.07 | No longer supported is replaced by CLK_N                     |
| 26  | CLK_1MS       |     | 16.12.07 | No longer supported is replaced by CLK_N                     |
| 26  | CLK_N         | 1.0 | 16.12.07 | New Module                                                   |
| 26  | EASTER        | 1.1 | 15.12.07 | changed code for better performance                          |
| 26  | REAL_TO_STRF  | 1.2 | 15.12.07 | changed code for better performance                          |
| 26  | DWORD_TO_STRH | 1.2 | 15.12.07 | changed code for better performance                          |
| 26  | DWORD_TO_STRB | 1.1 | 15.12.07 | changed code for better performance                          |
| 26  | BYTE_TO_STRH  | 1.2 | 15.12.07 | changed code for better performance                          |
| 26  | BYTE_TO_STRB  | 1.1 | 15.12.07 | changed code for better performance                          |
| 26  | FINDB         | 1.1 | 15.12.07 | changed code for better performance                          |
| 26  | TICKER        | 1.1 | 15.12.07 | step now starts at 1 instead of 0                            |
| 26  | _ARRAY_SORT   | 1.4 | 15.12.07 | added function return = true                                 |
| 26  | SIGNAL_4      | 1.0 | 13.12.07 | New Module                                                   |
| 26  | SIGNAL        | 1.0 | 13.12.07 | New Module                                                   |
| 26  | INT_TO_BCD    | 1.0 | 13.12.07 | New Module                                                   |
| 26  | BCD_TO_INT    | 1.0 | 13.12.07 | New Module                                                   |
| 26  | ARRAY_VAR     | 1.1 | 12.12.07 | changed code for better performance                          |
| 26  | ARRAY_AVG     | 1.1 | 12.12.07 | changed code for better performance                          |
| 26  | ARRAY_VAR     | 1.1 | 12.12.07 | changed code for better performance                          |
| 26  | ARRAY_TREND   | 1.1 | 12.12.07 | changed code for better performance                          |
| 26  | V3_ZANG       | 1.0 | 11.12.07 | New Module                                                   |
| 26  | V3_YANG       | 1.0 | 11.12.07 | New Module                                                   |
| 26  | V3_XANG       | 1.0 | 11.12.07 | New Module                                                   |
| 26  | V3_XPRO       | 1.0 | 11.12.07 | New Module                                                   |
| 26  | V3_SMUL       | 1.0 | 11.12.07 | New Module                                                   |
| 26  | V3_REV        | 1.0 | 11.12.07 | New Module                                                   |
| 26  | V3_PAR        | 1.0 | 11.12.07 | New Module                                                   |
| 26  | V3_NORM       | 1.0 | 11.12.07 | New Module                                                   |
| 26  | V3_DPRO       | 1.0 | 11.12.07 | New Module                                                   |
| 26  | V3_SUB        | 1.0 | 11.12.07 | New Module                                                   |
| 26  | V3_ADD        | 1.0 | 11.12.07 | New Module                                                   |
| 26  | V3_ANG        | 1.0 | 11.12.07 | New Module                                                   |
| 26  | V3_ABS        | 1.0 | 11.12.07 | New Module                                                   |
| 26  | LANGEVIN      | 1.0 | 10.12.07 | New Module                                                   |
| 26  | SIGMOID       | 1.0 | 10.12.07 | New Module                                                   |
| 26  | GAMMA         | 1.0 | 10.12.07 | New Module                                                   |
| 26  | FACT          | 1.2 | 10.12.07 | start value for i has changed to 2 for better performance    |
| 26  | TONOF         | 1.0 | 10.12.07 | New Module                                                   |
| 26  | PWM_PW        | 1.3 | 09.12.07 | Corrected a problem with F = 0                               |
| 252 | DCF77         | 1.6 | 08.12.07 | corrected a problem with time_zone_offset                    |
| 251 | LTIME_TO_UTC  | 1.3 | 08.12.07 | corrected a problem with time_zone_offset                    |
| 251 | LTIME         | 1.6 | 08.12.07 | corrected a problem with time_zone_offset                    |
| 251 | LOCAL_TIME    | 1.4 | 08.12.07 | corrected a problem with time_zone_offset                    |
| 25  | ESR_COLLECT   | 1.1 | 08.12.07 | Added reset input                                            |
| 25  | ESR_MON_R4    | 1.2 | 08.12.07 | corrected an error while esr type would not be set           |
| 25  | CHECK_PARITY  | 1.2 | 08.12.07 | changed code for better performance                          |
| 25  | PARITY        | 1.2 | 08.12.07 | changed code for better performance                          |
| 25  | WCT           | 1.1 | 07.12.07 | changed code for better performance                          |
| 25  | SPHERE_V      | 1.1 | 04.12.07 | changed code for better performance                          |
| 25  | CONE_V        | 1.1 | 04.12.07 | changed code for better performance                          |
| 25  | CIRCLE_A      | 1.1 | 04.12.07 | changed code for better performance                          |
| 25  | EXP_N         | 1.0 | 04.12.07 | New Module                                                   |
| 25  | TICKER        | 1.0 | 04.12.07 | New Module                                                   |
| 25  | AIR_DENSITY   | 1.1 | 04.12.07 | changed code for better performance                          |
| 25  | SQRTN         | 1.1 | 02.12.07 | changed code for better performance                          |
| 25  | ROUND         | 1.1 | 02.12.07 | changed code for better performance                          |
| 25  | ASINH         | 1.1 | 02.12.07 | changed code for better performance                          |
| 25  | ACOSH         | 1.1 | 02.12.07 | changed code for better performance                          |
| 25  | AOUT          | 1.3 | 02.12.07 | changed code for better performance                          |
| 25  | SDD           | 1.1 | 02.12.07 | changed code for better performance                          |
| 25  | DEW_TEMP      | 1.3 | 02.12.07 | changed code for better performance                          |



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|    |               |     |          |                                                                                                                       |
|----|---------------|-----|----------|-----------------------------------------------------------------------------------------------------------------------|
| 25 | DEW_CON       | 1.2 | 02.12.07 | changed code for better performance<br>output is now in Gramms / Qubic meter                                          |
| 25 | AIR_ENTHALPY  | 1.2 | 02.12.07 | changed code for better performance                                                                                   |
| 25 | EXP10         | 1.0 | 02.12.07 | New Module                                                                                                            |
| 25 | AIN           | 1.4 | 02.12.07 | chaged code for better performance                                                                                    |
| 25 | RMP_W         | 1.6 | 02.12.07 | corrected an error in calculation of step response                                                                    |
| 25 | TEMP_PT       | 1.3 | 02.12.07 | chaged code for better performance                                                                                    |
| 25 | SENSOR_INT    | 1.1 | 02.12.07 | Corrected an error in algorithm                                                                                       |
| 25 | RES_SI        | 1.1 | 02.12.07 | chaged code for better performance                                                                                    |
| 25 | RES_PT        | 1.2 | 02.12.07 | chaged code for better performance                                                                                    |
| 25 | RES_NI        | 1.2 | 02.12.07 | chaged code for better performance                                                                                    |
| 25 | ONTIME        | 1.4 | 02.12.07 | chaged code for better performance                                                                                    |
| 25 | BAR_GRAPH     | 1.1 | 02.12.07 | changed code to improove performance                                                                                  |
| 25 | FT_INT        | 1.5 | 02.12.07 | changed code to improove performance                                                                                  |
| 25 | WATER_DENSITY | 1.1 | 02.12.07 | new code for better performance                                                                                       |
| 25 | HEAT_INDEX    | 1.2 | 02.12.07 | new code for better performance                                                                                       |
| 25 | DEW_RH        | 1.1 | 02.12.07 | new code for better performance                                                                                       |
| 25 | DST           | 1.2 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | COTH          | 1.1 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | COSH          | 1.1 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | TANH          | 1.1 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | SINH          | 1.1 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | SIGN_R        | 1.2 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | SIGN_I        | 1.2 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | EVEN          | 1.1 | 01.12.07 | changed code to improove performance                                                                                  |
| 25 | FT_PT2        | 1.3 | 30.11.07 | changed out to be K * in during initialization<br>avoid divide by 0 if TN = 0                                         |
| 25 | FT_PT1        | 1.6 | 30.11.07 | changed out to be K * in during initialization                                                                        |
| 25 | PWM_PW        | 1.2 | 27.11.07 | avoid divide by 0 when F = 0                                                                                          |
| 25 | PWM_DC        | 1.2 | 27.11.07 | avoid divide by 0 when F = 0                                                                                          |
| 25 | GEN_SIN       | 1.2 | 27.11.07 | avoid divide by 0 when pt = 0                                                                                         |
| 25 | GEN_RMP       | 1.2 | 27.11.07 | avoid divide by 0 when pt = 0                                                                                         |
| 24 | BLIND_SHADE   | 1.2 | 24.11.07 | added shade_delay to avoid constant up and down if sun is out parcially                                               |
| 24 | HOLIDAY       | 1.3 | 24.11.07 | changes F_use of Buß_und_Bettag to 0 because this is no official holiday                                              |
| 24 | REAL_TO_STRF  | 1.1 | 20.11.07 | when N=0 there will be no dot at the end of the string                                                                |
| 24 | GEN_RDT       | 1.0 | 20.11.07 | New Module                                                                                                            |
| 24 | RDM           | 1.2 | 20.11.07 | changed code of temp calculation to avoid overflow in modr due to resuclt would not fit<br>DINT for high timer values |
| 24 | ACTUATOR_2P   | 1.4 | 19.11.07 | replaced left over statement time() with tx                                                                           |
| 24 | MIX           | 1.0 | 19.11.07 | New Module                                                                                                            |
| 24 | HEX_TO_DWORD  | 1.1 | 19.11.07 | changed type of function from int to dword                                                                            |
| 24 | HEAT_METER    | 1.0 | 17.11.07 | New Module                                                                                                            |
| 24 | _ARRAY_INIT   | 1.2 | 14.11.07 | changed size calculation to be more efficient                                                                         |
| 24 | _ARRAY_SORT   | 1.3 | 14.11.07 | changed size calculation to be more efficient                                                                         |
| 24 | _ARRAY_MEDIAN | 1.4 | 14.11.07 | corrected a problem with size calculation                                                                             |
| 24 | T_PLC_MS      | 1.1 | 12.11.07 | added temporaray variable tx because some compilers could not handle time() as an<br>argument                         |
| 24 | T_PLC_US      | 1.1 | 12.11.07 | added temporaray variable tx because some compilers could not handle time() as an<br>argument                         |
| 24 | LTIME_TO_UTC  | 1.2 | 12.11.07 | changed Type of time_zone_offset from time to int<br>to allow for time zones with negative offset                     |
| 24 | LTIME         | 1.5 | 12.11.07 | changed Type of time_zone_offset from time to int<br>to allow for time zones with negative offset                     |
| 24 | LOCAL_TIME    | 1.3 | 12.11.07 | changed time_zone_offset from time to integer to allow for negative offset time zones                                 |
| 24 | DCF77         | 1.5 | 12.11.07 | changed time_offset from time to integer to allow for negative offset time zones                                      |
| 24 | RDM           | 1.1 | 11.11.07 | changed time() into t_plc_ms()                                                                                        |
| 24 | MUL_ADD       | 1.1 | 11.11.07 | deleted preset values for K and O this makes no sense for a function                                                  |
| 24 | NEG           | 1.0 | 11.11.07 | New Module                                                                                                            |
| 24 | SGN           | 1.1 | 11.11.07 | changed type of function from real to int                                                                             |
| 24 | LTIME_TO_UTC  | 1.1 | 05.11.07 | replaced literal constant with variable because of error in möller ecp4 compiler                                      |
| 24 | CONTROL_SET2  | 1.0 | 04.11.07 | New Module                                                                                                            |
| 24 | CONTROL_SET1  | 1.0 | 04.11.07 | New Module                                                                                                            |
| 24 | TUNE          | 1.0 | 04.11.07 | New Module                                                                                                            |
| 24 | TUNE2         | 1.0 | 03.11.07 | New Module                                                                                                            |
| 24 | PARSET2       | 1.0 | 03.11.07 | New Module                                                                                                            |

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|    |                  |     |          |                                                                                                                                                   |
|----|------------------|-----|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 24 | FT_PI            | 1.1 | 03.11.07 | added noise input to filter noise<br>output diff added<br>set limit output false when output is within limits<br>overflow was not set correctly   |
| 24 | FT_PID           | 1.5 | 03.11.07 | added noise input to filter noise,<br>output diff added,<br>set limit output false when output is within limits<br>overflow was not set correctly |
| 24 | SET_TOD          | 1.3 | 02.11.07 | changed dword to DINT in calculation to avoid warnings with some compilers                                                                        |
| 24 | PARSET           | 1.0 | 02.11.07 | New Module                                                                                                                                        |
| 24 | DEAD_BAND        | 1.0 | 02.11.07 | New Module                                                                                                                                        |
| 24 | AIR_ENTHALPY     | 1.1 | 02.11.07 | deleted unused variable RF                                                                                                                        |
| 23 | DATE_ADD         | 1.2 | 01.11.07 | added int_to_dword statements to avoid possible overrun with möller ecp4                                                                          |
| 23 | SET_TOD          | 1.2 | 01.11.07 | changed coding to avoid possible overrun situation on möller ecp4                                                                                 |
| 23 | SET_DT           | 1.4 | 01.11.07 | converted hour type integer to dword in calculation to avoid overrun on möller ecp4                                                               |
| 23 | WINDOW2          | 1.0 | 31.10.07 | New Module                                                                                                                                        |
| 23 | FT_PID           | 1.4 | 31.10.07 | Rewritten to avoid error when one of the limits is 0                                                                                              |
| 23 | FT_PI            | 1.0 | 31.10.07 | New Module                                                                                                                                        |
| 23 | AIR_ENTHALPY     | 1.0 | 31.10.07 | New Module                                                                                                                                        |
| 23 | HOLIDAY          | 1.2 | 31.10.07 | changed holiday definition from constant to input constant                                                                                        |
| 23 | BLIND_CONTROL    | 1.3 | 30.10.07 | added lockout configuration time for motor direction change                                                                                       |
| 23 | AIR_DENSITY      | 1.0 | 29.10.07 | New Module                                                                                                                                        |
| 23 | SDD              | 1.0 | 29.10.07 | New Module                                                                                                                                        |
| 23 | WATER_ENTHALPY   | 1.0 | 29.10.07 | New Module                                                                                                                                        |
| 23 | FT_DERIV         | 1.3 | 29.10.07 | prohibit calculation when tx - last = 0 to avoid div 0 and increase accuracy on fast systems                                                      |
| 23 | FT_INT           | 1.4 | 29.10.07 | changed code so int will not be called if time difference is 0 to increase accuracy on systems with cycle times below 1ms                         |
| 23 | BLIND_SCENE      | 1.0 | 24.10.07 | new module                                                                                                                                        |
| 23 | LTIME            | 1.4 | 24.10.07 | changed code because the execution every 100ms can cause major problems<br>if the supplied time was not correct at start. use of new dst function |
| 23 | LOCAL_TIME       | 1.2 | 24.10.07 | changed dst calculation because function dst was upgraded no error in local_time only a change in DST                                             |
| 23 | DCF77            | 1.4 | 24.10.07 | changed dst calculation because function dst was upgraded no error in DCF77 only a change in DST                                                  |
| 23 | DST              | 1.1 | 24.10.07 | deleted time_zone_offset input because dst is generally at 01:00 utc and not mesz<br>uk starts 01:00 utc and also greece                          |
| 23 | FT_PT1           | 1.5 | 23.10.07 | added out := in to the init statements                                                                                                            |
| 23 | BLIND_SHADE      | 1.1 | 19.10.07 | manual mode was not detected because outputs were checked and not inputs UP and DN                                                                |
| 23 | CONE_V           | 1.0 | 17.10.07 | New Module                                                                                                                                        |
| 22 | GLOBAL CONSTANTS |     | 16.10.07 | Renamed PI in PI1 for better compatibility to other libraries                                                                                     |
| 22 | DEG              | 1.1 | 16.10.07 | Deg will become 0 when it reaches 360                                                                                                             |
| 22 | RAD              | 1.1 | 16.10.07 | Rad will become 0 when it reaches 2PI                                                                                                             |
| 22 | TRIANGLE_A       | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | SPHERE_V         | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | ELLIPSE_C        | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | ELLIPSE_A        | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | CIRCLE_C         | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | CIRCLE_A         | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | SGN              | 1.0 | 16.10.07 | New Module                                                                                                                                        |
| 22 | BLIND_CONTROL    | 1.2 | 16.10.07 | Ignore angle when T_ANGLE < T#100ms                                                                                                               |
| 22 | BLIND_SHADE      | 1.0 | 15.10.07 | New Module                                                                                                                                        |
| 22 | CRC_GEN          | 1.2 | 09.10.07 | Added init values for start value and final XOR<br>added reflect_in and reflect_out feature                                                       |
| 22 | REVERSE          | 1.0 | 09.10.07 | New Module                                                                                                                                        |
| 22 | SET_DT           | 1.3 | 08.10.07 | deleted unused variables count and leap                                                                                                           |
| 22 | DAY_OF_MONTH     | 1.3 | 08.10.07 | deleted unused variable yr                                                                                                                        |
| 22 | _ARRAY_MEDIAN    | 1.3 | 08.10.07 | deleted unused variables m and temp                                                                                                               |
| 22 | POLYNOM_INT      | 1.0 | 08.10.07 | New Module                                                                                                                                        |
| 22 | LINEAR_INT       | 1.0 | 07.10.07 | New Module                                                                                                                                        |
| 22 | BLIND_SECURITY   | 1.1 | 06.10.07 | Added pos and angle inputs and outputs<br>actions are now configurable                                                                            |
| 22 | BLIND_INPUT      | 1.1 | 06.10.07 | Renamed output ps and as to po and ao<br>added click mode                                                                                         |

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|----|------------------|-----|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22 | BLIND_NIGHT      | 1.1 | 06.10.07 | added pos and angle inputs and outputs<br>night position and angle can now be configured<br>manual operation at night will cancel night operation                                             |
| 22 | BLIND_CONTROL    | 1.1 | 06.10.07 | Integrated blind_actuator into BLIND_CONTROL                                                                                                                                                  |
| 22 | BLIND_ACTUATOR   | 1.1 | 06.10.07 | Rearranged outputs                                                                                                                                                                            |
| 22 | RMP_W            | 1.5 | 05.10.07 | Corrected a problem with pt = 0ms                                                                                                                                                             |
| 22 | RMP_B            | 1.5 | 05.10.07 | Corrected a problem with pt = 0ms                                                                                                                                                             |
| 22 | HEX_TO_DWORD     | 1.0 | 02.10.07 | New Module                                                                                                                                                                                    |
| 21 | DAY_OF_DATE      | 1.1 | 01.10.07 | Added step7 compatibility                                                                                                                                                                     |
| 21 | MONTH            | 1.1 | 01.10.07 | New coding for much better performance, no string functions used                                                                                                                              |
| 21 | SET_DT           | 1.2 | 01.10.07 | Added step7 compatibility                                                                                                                                                                     |
| 21 | SET_DATE         | 1.2 | 01.10.07 | added step7 compatibility, now call function set_date                                                                                                                                         |
| 21 | TIMER_1          | 1.5 | 01.10.07 | Corrected a problem , timer would not generate a pulse over midnight                                                                                                                          |
| 21 | DAY_OF_MONTH     | 1.2 | 01.10.07 | Corrected an error in algorithm.                                                                                                                                                              |
| 21 | LEAP_YEAR        | 1.2 | 01.10.07 | Improved code for higher performance                                                                                                                                                          |
| 21 | DAY_OF_YEAR      | 1.1 | 01.10.07 | Added compatibility to step7                                                                                                                                                                  |
| 21 | YEAR             | 1.1 | 01.10.07 | Corrected error and added compatibility to Step7                                                                                                                                              |
| 21 | GLOBAL CONSTANTS |     | 01.10.07 | Added Setup Parameter STEP7                                                                                                                                                                   |
| 21 | BLIND_NIGHT      | 1.0 | 30.09.07 | New Module                                                                                                                                                                                    |
| 21 | BLIND_SHADER     | 1.0 | 30.09.07 | New Module                                                                                                                                                                                    |
| 21 | BLIND_INPUT      | 1.0 | 29.09.07 | New Module                                                                                                                                                                                    |
| 21 | BLIND_CONTROL    | 1.0 | 29.09.07 | New Module                                                                                                                                                                                    |
| 21 | BLIND_SECURITY   | 1.0 | 29.09.07 | New Module                                                                                                                                                                                    |
| 21 | BLIND_ACTUATOR   | 1.0 | 29.09.07 | New Module                                                                                                                                                                                    |
| 21 | INTERLOCK        | 1.0 | 28.09.07 | New Module                                                                                                                                                                                    |
| 21 | RMP_W            | 1.4 | 28.09.07 | Added output high and low, coorrected error while Busy was displayed when en was false<br>improved performance and timing accuracy<br>added init statements to avoid wrong startup conditions |
| 21 | RMP_B            | 1.4 | 28.09.07 | Added output high and low, coorrected error while Busy was displayed when en was false<br>improved performance and timing accuracy<br>added init statements to avoid wrong startup conditions |
| 21 | _ARRAY_MEDIAN    | 1.2 | 22.09.07 | chagned code to use array_sort<br>Improved performance                                                                                                                                        |
| 21 | _ARRAY_SORT      | 1.2 | 22.09.07 | new algorithm for better performance                                                                                                                                                          |
| 21 | PERIOD           | 1.1 | 20.09.07 | corrected a problem with leap year                                                                                                                                                            |
| 20 | SET_DT           | 1.1 | 19.09.07 | Corrected leap year calculation                                                                                                                                                               |
| 20 | SET_DATE         | 1.1 | 19.09.07 | Corrected leap year calculation                                                                                                                                                               |
| 20 | PERIOD           | 1.0 | 19.09.07 | New Module                                                                                                                                                                                    |
| 20 | DCF77            | 1.3 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | ESR_MON_X8       | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | ESR_MON_R4       | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | ESR_MON_B8       | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | FT_RMP           | 1.3 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | TP_R             | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | SEQUENCE_8       | 1.4 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | SEQUENCE_4       | 1.4 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | GEN_SQ           | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | CLK_PULSE        | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | CLK_PRG          | 1.2 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | CLK_8MS          | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | CLK_4MS          | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | CLK_2MS          | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | CLK_1MS          | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | SH_2             | 1.3 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | SH_1             | 1.2 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | RMP_W            | 1.3 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | RMP_SOFT         | 1.2 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | RMP_B            | 1.3 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | GEN_SQ           | 1.2 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | GEN_SIN          | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | GEN_RMP          | 1.1 | 17.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                                                                                                       |
| 20 | DAYS_DELTA       | 1.1 | 16.09.07 | Corrected an error in algorithm                                                                                                                                                               |
| 20 | DAY_OF_DATE      | 1.0 | 16.09.07 | New Module                                                                                                                                                                                    |
| 20 | METER_STAT       | 1.0 | 16.09.07 | New Module                                                                                                                                                                                    |
| 20 | METER            | 1.0 | 16.09.07 | New Module                                                                                                                                                                                    |

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|    |               |     |          |                                                                                                         |
|----|---------------|-----|----------|---------------------------------------------------------------------------------------------------------|
| 20 | GEN_RDM       | 1.1 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | ONTIME        | 1.3 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | M_TX          | 1.2 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | M_T           | 1.2 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | M_D           | 1.1 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | FT_TN64       | 1.1 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | FT_TN16       | 1.1 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | FT_PT2        | 1.2 | 16.09.07 | Deleted unused code section for system time reading and unnecessary startup code                        |
| 20 | FT_PT1        | 1.4 | 16.09.07 | Replaced Time() with T_PLC_US for compatibility and better internal resolution                          |
| 20 | CYCLE_TIME    | 1.2 | 16.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | FT_INT        | 1.3 | 15.09.07 | Replaced Time() with T_PLC_US for compatibility and better internal resolution                          |
| 20 | FT_DERIV      | 1.2 | 15.09.07 | Replaced Time() with T_PLC_US for compatibility and better internal resolution                          |
| 20 | FT_PROFILE    | 1.1 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | TEMP_EXT      | 1.1 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | TIMER_2       | 1.2 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | TIMER_1       | 1.4 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | SWITCH_I      | 1.1 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | PULSE_LENGTH  | 1.1 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | DIMM_I        | 1.3 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | CLICK         | 1.2 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | ACTUATOR_UD   | 1.2 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | ACTUATOR_PUMP | 1.2 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | ACTUATOR_3P   | 1.3 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | ACTUATOR_2P   | 1.3 | 15.09.07 | Replaced Time() with T_PLC_MS for compatibility and performance reasons                                 |
| 20 | SHL1          | 1.0 | 14.09.07 | New Module                                                                                              |
| 20 | SHR1          | 1.0 | 14.09.07 | New Module                                                                                              |
| 20 | TOGGLE        | 1.0 | 13.09.07 | New Module                                                                                              |
| 20 | T_PLC_US      | 1.0 | 14.09.07 | New Module                                                                                              |
| 20 | T_PLC_MS      | 1.0 | 14.09.07 | New Module                                                                                              |
| 20 | AOUT          | 1.2 | 13.09.07 | changed code to avoid warning under codesys 2.3.8.1                                                     |
| 20 | AIN           | 1.3 | 13.09.07 | changed code to avoid warning under codesys 2.3.8.1                                                     |
| 20 | RES_NI        | 1.1 | 13.09.07 | Changed coding for better performance                                                                   |
| 20 | DEW_TEMP      | 1.2 | 13.09.07 | Changed algorithm to improve performance                                                                |
| 20 | DEW_CON       | 1.1 | 13.09.07 | Changed algorithm to improve performance                                                                |
| 20 | WATER_DENSITY | 1.0 | 13.09.07 | New Module                                                                                              |
| 20 | DWORD_TO_STRH | 1.1 | 11.09.07 | changed coding for compatibility with twincat, under twincat concat cannot have a function as argument. |
| 20 | BYTE_TO_STRH  | 1.1 | 11.09.07 | changed coding for compatibility with twincat, under twincat concat cannot have a function as argument. |
| 20 | SET_TOD       | 1.1 | 11.09.07 | changed coding to avoid a compiler warning under twincat.                                               |
| 20 | MATRIX        | 1.1 | 11.09.07 | deleted unused variables k and old_code                                                                 |
| 20 | CRC_GEN       | 1.1 | 11.09.07 | Deleted unused variable i                                                                               |
| 20 | CHECK_PARITY  | 1.1 | 10.09.07 | Changed algorithm to improve performance                                                                |
| 20 | PARITY        | 1.1 | 10.09.07 | Changed algorithm to improve performance                                                                |
| 20 | TEMP_SII      | 1.2 | 10.09.07 | Reduced accuracy to 0.01 degree to shorten execution time and corrected an error in formula             |
| 20 | TEMP_PT       | 1.2 | 10.09.07 | Reduced accuracy to 0.01 degree to shorten execution time                                               |
| 20 | TEMP_NI       | 1.2 | 10.09.07 | Reduced accuracy to 0.01 degree to shorten execution time                                               |
| 20 | BIT_COUNT     | 1.1 | 10.09.07 | Improved performance                                                                                    |
| 20 | SH_2          | 1.2 | 10.09.07 | Corrected an error while index would be out of array if N = 0;                                          |
| 20 | D_TRIG        | 1.0 | 04.09.07 | New Module                                                                                              |
| 18 | ONTIME        | 1.2 | 02.08.07 | Corrected an error while ontime would stop counting time after a certain time is reached                |
| 17 | LTIME_TO_UTC  | 1.0 | 05.07.07 | New Module                                                                                              |
| 17 | BIT_COUNT     | 1.0 | 05.07.07 | New Module                                                                                              |
| 17 | MATRIX        | 1.0 | 10.06.07 | New Module                                                                                              |
| 17 | CRC_CHECK     | 1.0 | 09.06.07 | New Module                                                                                              |
| 17 | CRC_GEN       | 1.0 | 09.06.07 | New Module                                                                                              |
| 17 | BYTE_TO_STRB  | 1.0 | 09.06.07 | New Module                                                                                              |
| 17 | BYTE_TO_STRH  | 1.0 | 09.06.07 | New Module                                                                                              |
| 17 | DWORD_TO_STRB | 1.0 | 09.06.07 | New Module                                                                                              |
| 17 | DWORD_TO_STRH | 1.0 | 09.06.07 | New Module                                                                                              |
| 17 | CLICK         | 1.1 | 18.05.07 | Corrected an error while Click would stop working when t_reconfig expired                               |
| 16 | PRESSURE      | 1.0 | 27.03.07 | New Module                                                                                              |
| 16 | ARRAY_SDV     | 1.0 | 01.04.07 | New Module                                                                                              |
| 16 | LTIME         | 1.3 | 18.03.07 | changed code, dst would not work during first cycle.                                                    |

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|    |                  |     |          |                                                                                                                                                                                            |
|----|------------------|-----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16 | DEW_TEMP         | 1.1 | 29.04.07 | Changed code when RH = 0, not to execute log(0)                                                                                                                                            |
| 16 | GLOBAL CONSTANTS |     | 02.05.07 | Added UG, universal gas constant                                                                                                                                                           |
| 16 | ARRAY_VAR        | 1.0 | 01.04.07 | New Module                                                                                                                                                                                 |
| 16 | TIMER_1          | 1.3 | 14.04.07 | Corrected an error while output would not go active when start year = 1970<br>increased accuracy by checking every 100ms<br>added init section to compensate for timer overflow at startup |
| 16 | DATE_ADD         | 1.1 | 12.04.07 | corrected an error while date would be incorrect when year = 0                                                                                                                             |
| 16 | M_TX             | 1.1 | 02.05.07 | Corrected wrong measurements during startup<br>added output ET (elapsed time)<br>added Reset input<br>added tmax input                                                                     |
| 16 | LENGTH           | 1.0 | 27.03.07 | New Module                                                                                                                                                                                 |
| 16 | ASTRO            | 1.0 | 27.03.07 | New Module                                                                                                                                                                                 |
| 16 | ENERGY           | 1.0 | 27.03.07 | New Module                                                                                                                                                                                 |
| 16 | DEW_CON          | 1.0 | 29.04.07 | New Module                                                                                                                                                                                 |
| 16 | CYCLE_TIME       | 1.2 | 10.03.07 | changed initialization of ct_min to t#10h for compatibility with siemens s7                                                                                                                |
| 16 | SPEED            | 1.0 | 27.03.07 | New Module                                                                                                                                                                                 |
| 16 | M_D              | 1.0 | 02.05.07 | New Module                                                                                                                                                                                 |
| 16 | M_T              | 1.1 | 02.05.07 | Added init variable to avoid unreasonable result if falling edge occurs first<br>added ET (elapsed Time) output<br>added reset input                                                       |
| 16 | DEW_RH           | 1.0 | 29.04.07 | New Module                                                                                                                                                                                 |
| 16 | SUN_TIME         | 1.2 | 17.04.07 | corrected error while sun:midday would not be corrected for longitude.                                                                                                                     |
| 15 | MUL_ADD          | 1.0 | 08.02.07 | New Module                                                                                                                                                                                 |
| 15 | CLK_PRG          | 1.1 | 25.02.07 | rewritten module in ST for better performance<br>PT can now be changed during runtime                                                                                                      |
| 15 | DEC_8            | 1.1 | 03.03.07 | rewritten module in ST for better performance                                                                                                                                              |
| 15 | FT_RMP           | 1.2 | 12.02.07 | Added INIT to prevent erroneous startup conditions                                                                                                                                         |
| 15 | GEN_RDM          | 1.0 | 07.02.07 | New Module                                                                                                                                                                                 |
| 15 | FT_PID           | 1.3 | 03.03.07 | Added default values for inputs KP, TN, TV, LIMIT_H and LIMIT_L                                                                                                                            |
| 15 | CEIL             | 1.0 | 07.02.07 | New Module                                                                                                                                                                                 |
| 15 | FT_PT1           | 1.3 | 27.02.07 | Output will be set to input during initial startup                                                                                                                                         |
| 15 | CLICK            | 1.0 | 15.02.07 | New Module „click“                                                                                                                                                                         |
| 15 | PULSE_LENGTH     | 1.0 | 15.02.07 | New Module                                                                                                                                                                                 |
| 15 | FACT             | 1.1 | 04.03.07 | Changed a critical error while loop was indefinite                                                                                                                                         |
| 15 | GEN_RMP          | 1.0 | 03.03.07 | New Module                                                                                                                                                                                 |
| 15 | FLOOR            | 1.0 | 07.02.07 | New Module                                                                                                                                                                                 |
| 15 | DEAD_ZONE_2      | 1.0 | 12.02.07 | New Module                                                                                                                                                                                 |
| 15 | SIGN_I           | 1.1 | 03.03.07 | changed method of function for better compatibility to other systems                                                                                                                       |
| 15 | GLOBAL CONSTANTS |     | 03.03.07 | Increased accuracy of PI2<br>renamed T0 to TK for better compatibility with other Libraries<br>renamed C to C0 for better compatibility                                                    |
| 15 | WCT              | 1.0 | 07.02.07 | New Module                                                                                                                                                                                 |
| 15 | HOLIDAYEAT_INDEX | 1.1 | 27.02.07 | deleted unused variables rh3 and T3                                                                                                                                                        |
| 15 | PWM_DC           | 1.1 | 25.02.07 | rewritten module in ST for better performance                                                                                                                                              |
| 15 | F_LAMP           | 1.1 | 08.02.07 | New Module                                                                                                                                                                                 |
| 15 | HOLIDAY          | 1.1 | 27.02.07 | deleted unused variable init                                                                                                                                                               |
| 15 | ONTIME           | 1.1 | 22.02.07 | Changed VAR RETAIN PERSISTENT to VAR RETAIN for better compatibility with<br>controllers without file system like WAGO's LON controllers                                                   |
| 15 | TIMER_2          | 1.1 | 22.02.07 | Changed VAR RETAIN PERSISTENT to VAR RETAIN for better compatibility with<br>controllers without file system like WAGO's LON controllers                                                   |
| 15 | _ARRAY_MEDIAN    | 1.1 | 03.03.07 | Corrected an error in median calculation                                                                                                                                                   |
| 15 | DEC_4            | 1.1 | 03.03.07 | rewritten module in ST for better performance                                                                                                                                              |
| 15 | GEN_SQR          | 1.1 | 12.02.07 | Added default value for dc = 0.5                                                                                                                                                           |
| 15 | SWITCH_X         | 1.0 | 15.02.07 | New Module                                                                                                                                                                                 |
| 15 | CALIBRATE        | 1.2 | 22.02.07 | Changed VAR RETAIN PERSISTENT to VAR RETAIN for better compatibility with<br>controllers without file system                                                                               |
| 15 | DEC_2            | 1.1 | 03.03.07 | rewritten module in ST for better performance                                                                                                                                              |
| 15 | FT_PROFILE       | 1.0 | 27.02.07 | New Module                                                                                                                                                                                 |
| 15 | INC              | 1.0 | 07.02.07 | New Module                                                                                                                                                                                 |
| 15 | PWM_PW           | 1.1 | 25.02.07 | rewritten module in ST for better performance                                                                                                                                              |
| 15 | DEAD_ZONE        | 1.0 | 12.02.07 | New Module                                                                                                                                                                                 |
| 15 | BAR_GRAPH        | 1.0 | 22.02.07 | New Module                                                                                                                                                                                 |
| 15 | BOILER           | 1.0 | 27.02.07 | New Module                                                                                                                                                                                 |
| 15 | DCF_77           | 1.2 | 26.02.07 | Changed statements to avoid a compiler error if wago 16 bit lon controller is used                                                                                                         |
| 14 | MUX_R2           | 1.0 | 19.01.07 | New Module                                                                                                                                                                                 |

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| 14 | DIMM_I        | 1.1 | 22.01.07 | Deleted unused structure „ramp“                                                                                                                                                                                                     |
| 14 | F_TO_OM       | 1.0 | 22.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | ESR_MON_X8    | 1.0 | 26.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | ASINH         | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | DAY_OF_MONTH  | 1.1 | 22.01.07 | deleted unused variable day_in_year and day_in_year_begin                                                                                                                                                                           |
| 14 | Heat_Temp     | 1.0 | 27.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | DIMM_I        | 1.2 | 02.02.07 | Changed method to read time() only at the beginning of the module for better consistency when a higher priority task interrupts „dimm_i“<br>added code in case structure for state 1 to allow for short one cycle long input pulses |
| 14 | SEQUENCE_4    | 1.3 | 17.01.07 | Changed output fault to ESR compatible output STATUS<br>added stop on error functionality<br>default for inputs IN is now TRUE<br>renamed variable state to step                                                                    |
| 14 | SH_2          | 1.1 | 20.01.07 | Added input N to specify the amount of samples for calculations<br>added output trig                                                                                                                                                |
| 14 | DWORD_TO_STRF | 1.0 | 16.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | _ARRAY_INIT   | 1.1 | 06.01.07 | Changed type of function to BOOL<br>function now returns TRUE                                                                                                                                                                       |
| 14 | SUN_POS       | 1.0 | 01.02.07 | New Module                                                                                                                                                                                                                          |
| 14 | SCALE_B4      | 1.1 | 19.01.07 | Changed outputs to REAL to avoid Integer overflow<br>added input offset for better cascading<br>changed from FB to FUNCTION                                                                                                         |
| 14 | SCALE_X8      | 1.0 | 19.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | STAIR2        | 1.0 | 28.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | HYST_3        | 1.0 | 22.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | GEN_SIN       | 1.0 | 22.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | SIGN_R        | 1.1 | 19.01.07 | changed method of function for better compatibility to other systems                                                                                                                                                                |
| 14 | RMP_W         | 1.2 | 22.01.07 | Deleted unused variable step                                                                                                                                                                                                        |
| 14 | KMH_TO_MS     | 1.0 | 04.02.07 | New Module                                                                                                                                                                                                                          |
| 14 | ACTUATOR_2P   | 1.2 | 17.01.07 | Deleted unused variable time_overflow                                                                                                                                                                                               |
| 14 | ACTUATOR_3P   | 1.2 | 23.01.07 | Deleted unused variables „force_on“ and „force_off“                                                                                                                                                                                 |
| 14 | FT_PT2        | 1.1 | 15.01.07 | Improved algorithm for higher accuracy and speed                                                                                                                                                                                    |
| 14 | SENSOR_INT    | 1.1 | 20.01.07 | Deleted input R0 which was not used                                                                                                                                                                                                 |
| 14 | SQRTN         | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | BURNER        | 1.3 | 31.12.06 | Added status output to be ESR compatible                                                                                                                                                                                            |
| 14 | F_LIN2        | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | FT_TN16       | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | FT_PID        | 1.2 | 03.01.07 | Added int_band for structural adaptation of PID                                                                                                                                                                                     |
| 14 | BYTE_OF_DWORD | 1.0 | 17.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | AOUT          | 1.1 | 18.01.07 | Renamed module to AOUT<br>changed output to hold 32 bits max<br>corrected an error with sign bit                                                                                                                                    |
| 14 | SUN_TIME      | 1.1 | 20.01.07 | deleted unused variables sun_rineR and sun_setR                                                                                                                                                                                     |
| 14 | SH            | 1.1 | 16.01.07 | Added output trig                                                                                                                                                                                                                   |
| 14 | MAX_3         | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | OM_TO_F       | 1.0 | 22.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | MIN_3         | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | FT_PT1        | 1.2 | 03.01.07 | Corrected an error with falling edge errors<br>added faktor K                                                                                                                                                                       |
| 14 | SEQUENCE_8    | 1.3 | 17.01.07 | Changed output fault to ESR compatible output STATUS<br>added stop on error functionality<br>default for inputs IN is now TRUE<br>renamed variable state to step                                                                    |
| 14 | MULTI_IN      | 1.1 | 01.01.07 | Changed midr to mid3 funktion for better performance                                                                                                                                                                                |
| 14 | STAIR         | 1.0 | 28.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | ACOSH         | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | BURNER        | 1.2 | 31.12.06 | Added Oil_Temp input                                                                                                                                                                                                                |
| 14 | ACTUATOR_UD   | 1.1 | 22.01.07 | Added status output<br>changed inputs „ton“ and „toff“ to config variables<br>added config var „out_return to enable „yup_in“ and y_dn_in“<br>changed manual mode to override automatic mode                                        |
| 14 | FT_TN8        | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                                          |
| 14 | SCALE_B8      | 1.1 | 19.01.07 | Changed outputs to REAL to avoid Integer overflow<br>added input offset for better cascading<br>changed from FB to FUNCTION                                                                                                         |
| 14 | DELAY         | 1.2 | 19.01.07 | changed reset to load the value of in instead of 0                                                                                                                                                                                  |
| 14 | CHR           | 1.0 | 16.01.07 | New Module                                                                                                                                                                                                                          |

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| 14 | SCALE_X4      | 1.0 | 19.01.07 | New Module                                                                                                                                                                                                  |
| 14 | RDM           | 1.0 | 16.01.07 | New Module                                                                                                                                                                                                  |
| 14 | LEGIONELLA    | 1.1 | 17.01.07 | Rewritten module in ST for better performance<br>changed error out to ESR compatible Status output                                                                                                          |
| 14 | ACOTH         | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | WORK_WEEK     | 1.1 | 14.01.07 | deleted unused variable yday                                                                                                                                                                                |
| 14 | SCALE_X2      | 1.0 | 19.01.07 | New Module                                                                                                                                                                                                  |
| 14 | DCF_77        | 1.1 | 02.02.07 | change wday and dst outputs when there is no dcf reception                                                                                                                                                  |
| 14 | FT_IMP        | 1.1 | 03.01.07 | Added faktor K                                                                                                                                                                                              |
| 14 | SH_1          | 1.1 | 06.01.07 | Added output trig                                                                                                                                                                                           |
| 14 | MUX_R4        | 1.0 | 19.01.07 | New Module                                                                                                                                                                                                  |
| 14 | DELAY_4       | 1.1 | 19.01.07 | added variable temp to add 1 delay for q1                                                                                                                                                                   |
| 14 | WORD_OF_DWORD | 1.0 | 17.01.07 | New Module                                                                                                                                                                                                  |
| 14 | _ARRAY_SORT   | 1.1 | 06.01.07 | Function now returns TRUE                                                                                                                                                                                   |
| 14 | BURNER        | 1.4 | 22.01.07 | Changed output „fail“ to „status“<br>added output „cycles“<br>corrected an error while pre_heat would be active when an error was present<br>added „coil2“ output and „stage2“ input for dual stage burners |
| 14 | TEMP_PT       | 1.1 | 24.01.07 | deleted input R0 which was not used                                                                                                                                                                         |
| 14 | OFFSET_2      | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | FT_PT1        | 1.1 | 01.01.07 | Corrected error during startup<br>corrected error with very small time values < 10ms                                                                                                                        |
| 14 | COSH          | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | TANH          | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | FT_DERIV      | 1.1 | 03.01.07 | Added init code for startup<br>set default for K to 1                                                                                                                                                       |
| 14 | RMP_SOFT      | 1.0 | 22.01.07 | New Module                                                                                                                                                                                                  |
| 14 | ATANH         | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | FT_INT        | 1.2 | 15.01.07 | Added default value for K = 1                                                                                                                                                                               |
| 14 | COTH          | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | DWORD_OF_BYTE | 1.0 | 24.01.07 | New Module                                                                                                                                                                                                  |
| 14 | MS_TO_KMH     | 1.0 | 04.02.07 | New Module                                                                                                                                                                                                  |
| 14 | RMP_B         | 1.2 | 17.01.07 | Renamed input UD to UP for better usability<br>deleted unused variable step                                                                                                                                 |
| 14 | TEMP_NI       | 1.1 | 24.01.07 | deleted input R0 which was not used                                                                                                                                                                         |
| 14 | ESR_MON_B8    | 1.0 | 26.01.07 | New Module                                                                                                                                                                                                  |
| 14 | ESR_MON_R4    | 1.0 | 26.01.07 | New Module                                                                                                                                                                                                  |
| 14 | FT_AVG        | 1.1 | 07.01.07 | Changed reset logic to load buffer with IN instead of 0<br>added EN input for better flow control<br>added init to avoid erroneous startup conditions<br>deleted unused variable CNT                        |
| 14 | OFFSET        | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | EXEC          | 1.1 | 06.02.07 | Corrected error with COS function<br>divide by 0 will return an error now                                                                                                                                   |
| 14 | AIN           | 1.2 | 19.01.07 | fixed an error with sign bit.                                                                                                                                                                               |
| 14 | LTIME         | 1.2 | 17.01.07 | Added input UTC instead of internal SYSRTCGETTIME for better compatibility with different Hardware<br>corrected an error while DST_ENABLE would not be checked before DST is enabled                        |
| 14 | BURNER        | 1.1 | 31.12.06 | Added Pre heating time and output<br>flame monitor before ignition will generate error output now<br>added output for Kwh                                                                                   |
| 14 | WINDOW        | 1.1 | 22.01.07 | Changed input arrangement                                                                                                                                                                                   |
| 14 | TEMP_SI       | 1.1 | 24.01.07 | deleted input R0 which was not used                                                                                                                                                                         |
| 14 | WORD_OF_BYTE  | 1.0 | 24.01.07 | New Module                                                                                                                                                                                                  |
| 14 | ESR_COLLECT   | 1.0 | 26.01.07 | New Module                                                                                                                                                                                                  |
| 14 | SINH          | 1.0 | 12.01.07 | New Module                                                                                                                                                                                                  |
| 14 | FT_TN64       | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                  |
| 14 | MID_3         | 1.0 | 01.01.07 | New Module                                                                                                                                                                                                  |
| 14 | SCALE_B2      | 1.1 | 19.01.07 | Changed outputs to REAL to avoid Integer overflow<br>added input offset for better cascading<br>changed from FB to FUNCTION                                                                                 |
| 13 | ACTUATOR_2P   | 1.1 | 07.10.06 | changed error, pos could not reach 1 or 0 under certain conditions due to error in code                                                                                                                     |
| 13 | DATE_ADD      | 1.0 | 27.12.06 | New Module                                                                                                                                                                                                  |
| 13 | TEMP_EXT      | 1.0 | 06.12.06 | New Module                                                                                                                                                                                                  |
| 13 | K_TO_C        | 1.0 | 04.12.06 | New Module                                                                                                                                                                                                  |
| 13 | RMP_B         | 1.1 | 15.10.06 | Added output busy                                                                                                                                                                                           |

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| 13 | FIND_NONUM    | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | FT_PID        | 1.1 | 01.12.06 | Changed algorithm to „Trapezregel“ for higher accuracy                                                |
| 13 | TIMER_1       | 1.2 | 25.12.06 | Corrected an error with timer overflow                                                                |
| 13 | TIMER_1       | 1.1 | 01.12.06 | Corrected an error with pulse width = 0                                                               |
| 13 | C_TP_K        | 1.0 | 04.12.06 | New Module                                                                                            |
| 13 | CYCLE_TIME    | 1.1 | 12.12.06 | Added output cycles                                                                                   |
| 13 | TRIM          | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | FINDB_NUM     | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | MUX_4         | 1.1 | 05.10.06 | rewritten module in ST for better performance                                                         |
| 13 | FT_RMP        | 1.1 | 08.10.06 | Added UD output                                                                                       |
| 13 | FIND_NUM      | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | ACTUATOR_2P   | 1.1 | 07.10.06 | changed „on“ and „off“ to „force“ and „force_in“ for better usability.                                |
| 13 | SEQUENCE_4    | 1.2 | 01.12.06 | corrected an error in sequence logic<br>added init at startup to prevent erroneous startup conditions |
| 13 | MUX_2         | 1.1 | 05.10.06 | rewritten module in ST for better performance                                                         |
| 13 | UPPERCASE     | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | DAYS_DELTA    | 1.0 | 27.12.06 | New Module                                                                                            |
| 13 | FT_INT        | 1.1 | 13.12.06 | Changed algorithm to „Trapezregel“ for higher accuracy                                                |
| 13 | ACTUATOR_3P   | 1.1 | 19.10.06 | Added code to avoid overrun of the end_switch                                                         |
| 13 | OSCAT_VERSION | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | LOWERCASE     | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | FINDB         | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | REAL_TO_STRF  | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | ACTUATOR_PUMP | 1.1 | 27.12.06 | Fixed an error while the pump would run for t_min at startup                                          |
| 13 | EASTER        | 1.0 | 27.12.06 | New Module                                                                                            |
| 13 | DIFFER        | 1.0 | 08.10.06 | New Module                                                                                            |
| 13 | FINDB_NONUM   | 1.0 | 06.10.06 | New Module                                                                                            |
| 13 | SEQUENCE_8    | 1.2 | 01.12.06 | corrected an error in sequence logic<br>added init at startup to prevent erroneous startup conditions |
| 13 | STATUS_TO_ESR | 1.0 | 06.10.06 | New Module                                                                                            |
| 12 | SECOND        | 1.1 | 02.10.06 | Changed name of input to ITOD                                                                         |
| 12 | RMP_W         | 1.1 | 04.10.06 | added the busy output which signals that the ramp is running.                                         |
| 12 | ARRAY_SUM     | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | ARRAY_TREND   | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | LTIME         | 1.1 | 02.10.06 | corrected an error where dst would be delayed by 0.1second                                            |
| 12 | LEAP_OF_DATE  | 1.1 | 02.10.06 | Corrected an error while year 1970 would not work                                                     |
| 12 | CHECK_PARITY  | 1.0 | 01.10.06 | New Module                                                                                            |
| 12 | DELAY         | 1.1 | 01.10.06 | Corrected an error in buffer management                                                               |
| 12 | ARRAY_MIN     | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | ARRAY_MAX     | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | SEQUENCE_4    | 1.1 | 01.10.06 | Added output any<br>corrected delay to occur after event and not before                               |
| 12 | LEAP_YEAR     | 1.1 | 02.10.06 | Corrected an error while year 1970 would not work                                                     |
| 12 | ARRAY_AVG     | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | ARRAY_SPR     | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | LOCAL_TIME    | 1.1 | 02.10.06 | Corrected an error while DST would be calculated wrong                                                |
| 12 | HOURL         | 1.1 | 02.10.06 | Changed name of input to ITOD                                                                         |
| 12 | MULTIME       | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | EVEN          | 1.0 | 01.10.06 | New Module                                                                                            |
| 12 | DEW_TEMP      | 1.0 | 02.10.06 | New Module                                                                                            |
| 12 | MINUTE        | 1.1 | 02.10.06 | Changed name of input to ITOD                                                                         |
| 12 | SEQUENCE_8    | 1.1 | 01.10.06 | Added output any<br>corrected delay to occur after event and not before                               |
| 11 | SH_2          | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | SH_T          | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | FT_MIN_MAX    | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | PARITY        | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | F_LIN         | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | F_QUAD        | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | AIN           | 1.1 | 18.08.06 | fixed an error with low value negative and high value 0.                                              |
| 11 | ROUND         | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | F_POWER       | 1.0 | 01.09.06 | New Module                                                                                            |
| 11 | WEEKDAY       | 1.1 | 21.08.06 | Corrected an error in formula                                                                         |
| 10 | TP_R          | 1.0 | 04.08.06 | New Module                                                                                            |
| 10 | SET_DT        | 1.0 | 04.08.06 | New Module                                                                                            |



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| 10 | M_TX           | 1.0 | 04.08.06 | New Module |
| 10 | RAD            | 1.0 | 04.08.06 | New Module |
| 10 | INC_DEC        | 1.0 | 04.08.06 | New Module |
| 10 | MINUTE_TO_TIME | 1.0 | 04.08.06 | New Module |
| 10 | SHR_4UDE       | 1.0 | 04.08.06 | New Module |
| 10 | DAY_OF_YEAR    | 1.0 | 04.08.06 | New Module |
| 10 | HYST_2         | 1.0 | 04.08.06 | New Module |
| 10 | MUX_4          | 1.0 | 04.08.06 | New Module |
| 10 | SECOND_TO_TIME | 1.0 | 04.08.06 | New Module |
| 10 | FF_D2E         | 1.0 | 04.08.06 | New Module |
| 10 | CLK_4MS        | 1.0 | 04.08.06 | New Module |
| 10 | DST            | 1.0 | 04.08.06 | New Module |
| 10 | FRACT          | 1.0 | 04.08.06 | New Module |
| 10 | SET_TOD        | 1.0 | 04.08.06 | New Module |
| 10 | FF_D4E         | 1.0 | 04.08.06 | New Module |
| 10 | CLK_DIV        | 1.0 | 04.08.06 | New Module |
| 10 | CLK_2MS        | 1.0 | 04.08.06 | New Module |
| 10 | PT_TO_F        | 1.0 | 04.08.06 | New Module |
| 10 | CLK_PULSE      | 1.0 | 04.08.06 | New Module |
| 10 | M_T            | 1.0 | 04.08.06 | New Module |
| 10 | DAY_OF_MONTH   | 1.0 | 04.08.06 | New Module |
| 10 | DAY_TO_TIME    | 1.0 | 04.08.06 | New Module |
| 10 | RES_PT         | 1.1 | 04.08.06 | New Module |
| 10 | FIFO_16        | 1.0 | 04.08.06 | New Module |
| 10 | STACK_32       | 1.0 | 04.08.06 | New Module |
| 10 | F_TO_PT        | 1.0 | 04.08.06 | New Module |
| 10 | RES_NI         | 1.0 | 04.08.06 | New Module |
| 10 | CLK_1MS        | 1.0 | 04.08.06 | New Module |
| 10 | MUX_2          | 1.0 | 04.08.06 | New Module |
| 10 | BIT_OF_DWORD   | 1.0 | 04.08.06 | New Module |
| 10 | FF_JKE         | 1.0 | 04.08.06 | New Module |
| 10 | LTCH_4         | 1.0 | 04.08.06 | New Module |
| 10 | GEN_SQ         | 1.0 | 04.08.06 | New Module |
| 10 | GEN_BIT        | 1.0 | 04.08.06 | New Module |
| 10 | SHR_8UDE       | 1.0 | 04.08.06 | New Module |
| 10 | HOURL          | 1.0 | 04.08.06 | New Module |
| 10 | SHR_4E         | 1.0 | 04.08.06 | New Module |
| 10 | B_TRIG         | 1.0 | 04.08.06 | New Module |
| 10 | BYTE_OF_BIT    | 1.0 | 04.08.06 | New Module |
| 10 | BYTE_TO_BIT    | 1.0 | 04.08.06 | New Module |
| 10 | SHR_8PLE       | 1.0 | 04.08.06 | New Module |
| 10 | C_TO_F         | 1.0 | 04.08.06 | New Module |
| 10 | FF_DRE         | 1.0 | 04.08.06 | New Module |
| 10 | STACK_16       | 1.0 | 04.08.06 | New Module |
| 10 | SET_DATE       | 1.0 | 04.08.06 | New Module |
| 10 | DEG            | 1.0 | 04.08.06 | New Module |
| 10 | SWITCH_I       | 1.0 | 04.08.06 | New Module |
| 10 | HOURL_TO_TOD   | 1.0 | 04.08.06 | New Module |
| 10 | YEAR           | 1.0 | 04.08.06 | New Module |
| 10 | HYST_1         | 1.0 | 04.08.06 | New Module |
| 10 | MONTH          | 1.0 | 04.08.06 | New Module |
| 10 | HOURL_TO_TIME  | 1.0 | 04.08.06 | New Module |
| 10 | LTCH           | 1.0 | 04.08.06 | New Module |
| 10 | CLK_8MS        | 1.0 | 04.08.06 | New Module |
| 10 | RES_SI         | 1.0 | 04.08.06 | New Module |
| 10 | F_TO_C         | 1.0 | 04.08.06 | New Module |
| 10 | FIFO_32        | 1.0 | 04.08.06 | New Module |